

RentalPro

Digital Rental Management System

IT402 Capstone Project Defense · All Six Deliverables Complete + Billing Engine + Full Arabic + Production Hardening

Presented by: Abdurrahmaan Ali

Course: IT402 — IT Capstone II

The Problem

Independent landlords and small property managers still run their business on spreadsheets, paper leases, and manual reminders.

- 1 No single source of truth**
Rent, leases, and maintenance history live in different spreadsheets and email threads.
- 2 Missed rent and renewals**
Nothing proactively flags overdue rent or an expiring lease before it becomes a problem.
- 3 No accountability trail**
Verbal maintenance requests and cash payments leave no audit trail for disputes.
- 4 No self-service for tenants**
Tenants have to call or email for basics: a balance, a receipt, a repair update.

Target Users

- Property Owners
- Property Managers
- Tenants
- Maintenance Staff
- Administrators

“One system, five roles, zero spreadsheets.”

Project Objectives

1

Centralize property & lease data

One record per property, unit, tenant, and lease — no duplicate spreadsheets.

2

Automate the reminders that get missed

Background jobs alert on lease expiry, rent due, overdue rent, and overdue maintenance.

3

Give every role a role-scoped view

RBAC + ownership checks so an Owner never sees another Owner's portfolio.

4

Make every record auditable

Immutable payments and an append-only audit log — no silent edits.

5

Ship it fully tested, not just working

100% of SRS requirements implemented and covered by an automated test.

Requirements — IT401 Deliverable 1

The Software Requirements Specification defined 25 use cases across six functional areas.

Auth & RBAC

2 use cases

Login, role-based access for 5 roles

Property & Unit Mgmt

3 use cases

Add/edit properties, units, dashboard

Tenant Lifecycle

4 use cases

Onboarding, invites, leases, portal

Rent & Payments

6 use cases

Recording, statements, alerts, history

Maintenance

6 use cases

Submit through close, cost tracking

Reporting

2 use cases

Performance reports, PDF/JSON export

System Design — IT401 Deliverable 2

UML class model and a normalized relational schema — 13 tables, all linked by UUID foreign keys.



Design decisions carried through to the build

UUID primary keys (identical schema on SQLite and PostgreSQL) • Naive UTC datetimes • Soft-status fields, never physical deletes • Immutable payments • JSON audit diffs

Architecture

A layered Flask application — five layers, each with a single responsibility.

PRESENTATION

40 Jinja2 templates, role-aware top navigation, English/Shona/Arabic menu

APPLICATION

10 Blueprints — auth, properties, tenants, rent, maintenance, admin, reports, portal, api

DOMAIN / SERVICE

security.py (RBAC, IDOR), web_security.py (headers, secret key), i18n.py (English/Shona/Arabic), scheduler_jobs.py, services/ (pdf, excel, notifications, payments, storage)

DATA

18 SQLAlchemy models (13 designed + 5 billing), UUID keys, schema managed by Alembic migrations

INFRASTRUCTURE

PostgreSQL/SQLite · Redis · Docker Compose · local or S3-compatible file storage · dev-mode fallback for SendGrid/Twilio/Stripe

Key Features by Role

Owner	Manager	Tenant	Staff	Admin
Create & edit properties/units	Manage assigned properties	Self-service portal	Personal work queue only	User & role management
Configure late fees	Onboard & blacklist tenants	Pay rent online	Update ticket status	Create staff accounts
Full portfolio reports	Assign maintenance work	Submit requests with photos	Attach completion photos	Full data visibility

IT402 Deliverable 2 — 100% Feature Completion

Eleven gaps closed against the SRS after the IT401 Deliverable 3 prototype, grouped here into eight areas — every one covered by an automated test.



Photo uploads

Property (≤ 10), unit (≤ 5), and maintenance (≤ 3) photos — JPEG/PNG validated by magic bytes, not extension



Unit editing

Edit a unit's details and status after it is created — previously units could only be added



Configurable late fees

Fixed or percentage, applied automatically once rent is overdue



Tenant blacklisting

Flag a tenant to block future leases without touching an existing one



Tenant & lease documents

Upload a tenant's ID; download a lease PDF as management or as the tenant



Admin user console

Role & status management with an “at least one active Owner” guard



Staff self-service accounts

Owner/Manager/Admin can create a Maintenance Staff login directly



CSV & Excel exports

A lease's payment history as CSV; a two-sheet .xlsx portfolio workbook

Extended Scope — The Billing Engine

Eight further billing capabilities added after the first demo package, going beyond the original SRS — 79 additional tests, all passing.



Operational & sundry charges

One-off charges beyond rent — cleaning, call-outs — billed straight to the lease



Electricity metering

A per-property rate plus manual meter readings; consumption bills itself automatically



Pro rata rent

The first month prorates to a daily rate when a tenant moves in mid-month



Variable / escalating rent

A scheduled rent revision takes effect from a chosen date — past periods stay untouched



Automated recurring billing

A daily scheduler job invoices every elapsed period exactly once, with a full paper trail



Account statements by period

Opening balance, in-period activity, closing balance — plus full tenant self-service access



Payment receipts

Every payment now has a real, downloadable PDF receipt, not just a receipt number



Property expenses & net income

Owner-side operating costs feed a true net income estimate, not just rent collected

Extended Scope — Usability & Localisation

Built for Harare: a clearer interface, two more languages and realistic local data — 27 more tests, all passing.



Property map

Click-to-pin coordinates on every property, plus a portfolio map of all of them



Top-bar redesign

The sidebar became a horizontal top bar with a soft blue-to-burgundy gradient



Harare photography

A full-screen Harare skyline login, and a Harare photo behind each section of the app



English, Shona & Arabic

A language menu in the top bar and on login; Arabic reads right to left; the choice is remembered



Place names stay as entered

Only interface text is translated — a test fails if a Harare place name is ever altered



Tenant-friendly unit labels

Tenants see “Avondale Heights · Unit 101”, not an internal unit code



Logged-in redirect fix

Opening the login page while signed in now goes to your dashboard, not a blank page



Harare demo portfolio

5 properties in Avondale, the CBD, Eastlea and Vainona — 16 units and 9 tenants

Extended Scope — Full Arabic & Production Hardening

The final round: every word in Arabic, a stricter security posture and production-ready data handling — 54 more tests, all passing.



Arabic on every screen

Every page, form option and message — over 600 phrases; a test fails if one is missing



Emails & SMS in each user's language

Sent in the language the user last chose, including by the nightly jobs



Arabic PDFs & spreadsheets

PDF letters joined and right to left, tables mirrored; Excel sheets open right to left



Currency everywhere

Every amount shown as “USD 1,250.00” — screens, messages, emails and PDFs



Security headers & cookies

Strict Content-Security-Policy with no inline code, no framing, HSTS, HttpOnly cookies



No default secret key

The app refuses to start with a missing or published SECRET_KEY



Database migrations

Alembic replaces create_all(); older databases are adopted with their data

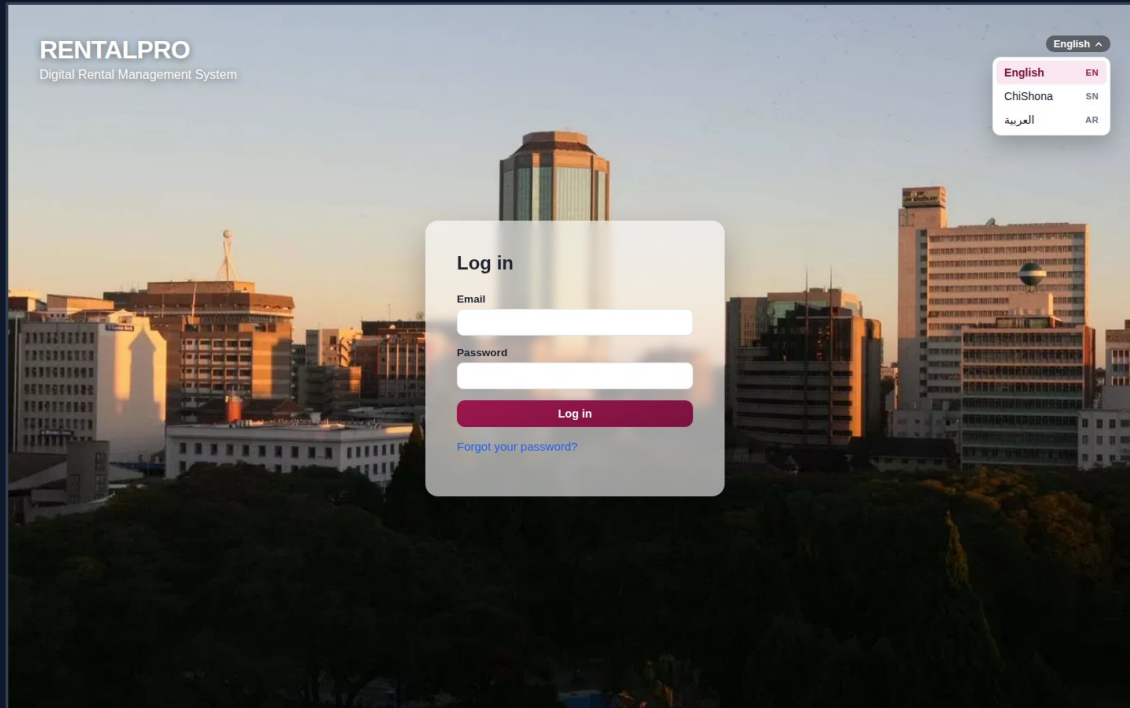


Cloud file storage

Uploads can live in an S3-compatible bucket and survive redeloys

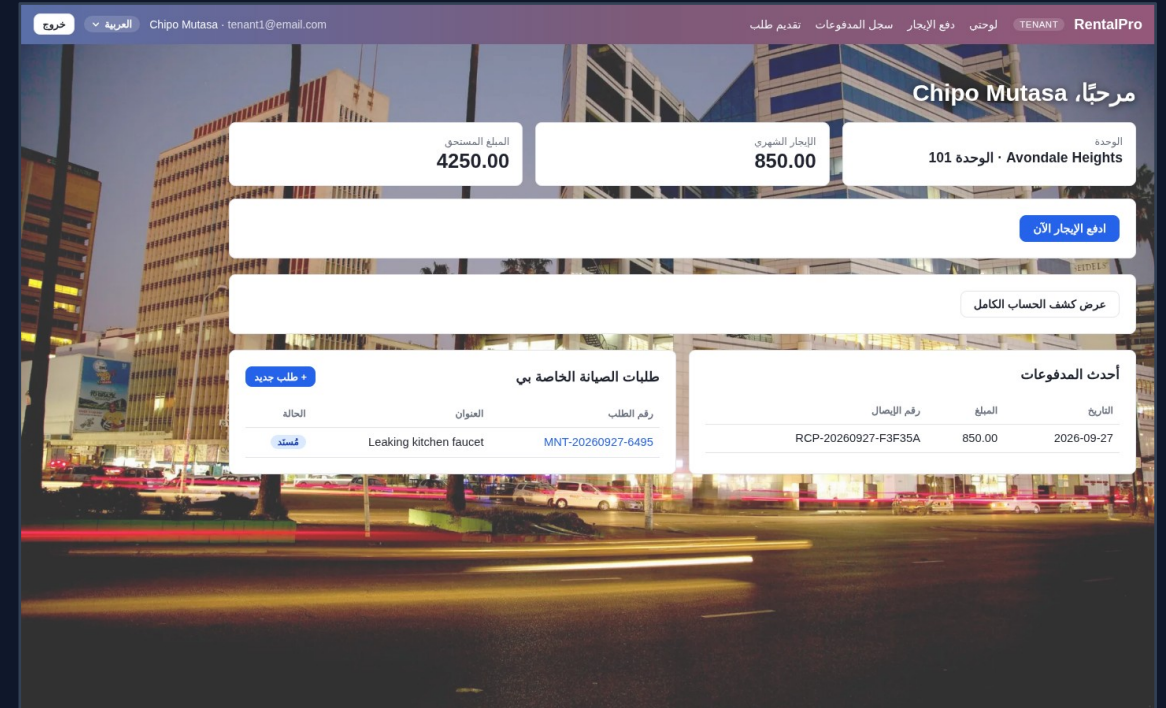
The New Look — Harare, Shona & Arabic

The Harare login page with its language menu, and the tenant portal in Arabic



Login

A full-screen Harare skyline, a translucent card, and the language menu open top-right



Tenant portal in Arabic

Laid out right to left — her name and “Avondale Heights” stay exactly as entered

Testing & Quality

305

Automated Tests

0

Failures

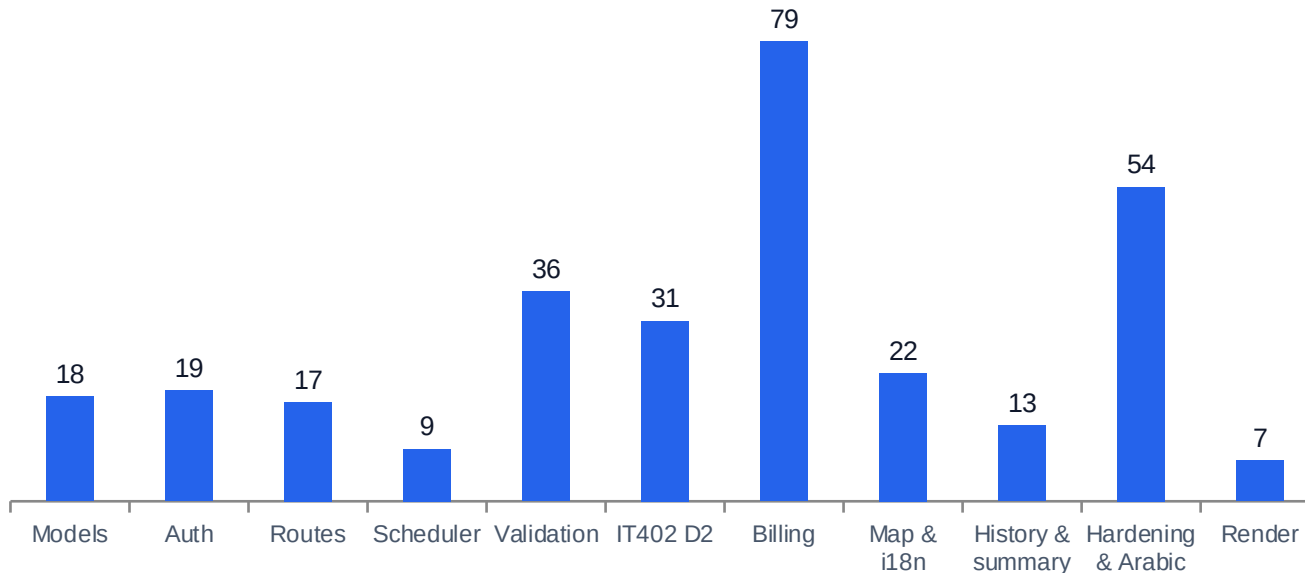
86%

Statement Coverage

25/25

Use Cases Covered

Tests per Module



What the tests actually verify

- Real HTTP requests through Flask's test client — not isolated function calls
- RBAC across all 5 roles × every management route
- IDOR: a second Owner cannot view the first Owner's property
- Calendar edge cases: mid-month leases, day-31-in-February
- Stripe HMAC signature verification — valid, tampered, expired

Live Demo

A walkthrough of the running application

- 1 Owner: property & unit setup, photos, and a configured late fee
- 2 Owner: onboard a tenant, upload a lease, activate the tenancy
- 3 Owner: blacklist / unblacklist, record a rent payment
- 4 Tenant → Owner → Staff → Tenant: full maintenance lifecycle
- 5 Admin: user management and staff account creation
- 6 Billing engine, then the Harare map and the language menu

Challenges & Solutions

CHALLENGE

Reconciling ambitious design docs with what one developer could actually build and verify

SOLUTION

Simplified IT401 Deliverable 2's microservice sketch to a layered monolith; corrected IT401 Deliverable 3's overstated early numbers to the real, verified ones

CHALLENGE

Calendar-accurate billing — a lease starting on the 31st, or a due day that doesn't exist in February

SOLUTION

Wrote the edge-case tests first (mid-month start, day-29 clipping) before trusting the due-date calculation

CHALLENGE

Testing payments, email, and SMS without any live paid account

SOLUTION

Built a dev-mode fallback pattern into every external service — the whole app runs and tests fully offline

Security Controls



Passwords & secret key

pbkdf2:sha256, 600,000 iterations; no start without a real SECRET_KEY



IDOR protection

assert_owner() / assert_tenant_self() — resource-level, not just route-level



Headers & lockout

Strict Content-Security-Policy, no framing, HSTS; 5 failed logins lock an account



Audit log

Immutable, append-only record of every state-changing action



RBAC

role_required() on every protected route, 5 roles



CSRF & cookies

Flask-WTF on every form (signed Stripe webhook exempt); HttpOnly, SameSite session cookie



Upload validation

Magic-byte checks — PDF, JPEG, PNG — regardless of file extension



Webhook verification

Manual HMAC check, 300-second replay window, independent of the SDK

Conclusion

RentalPro was carried from requirements through design, a working prototype, a full project report, 100% feature completion, a billing engine, and a fully Arabic, production-hardened Harare interface — every claim backed by a running, tested system.

- ✓ 25 / 25 use cases implemented, each with a route and a covering test
- ✓ 13-table database design implemented exactly as specified
- ✓ 305 automated tests, 0 failures, 86% statement coverage
- ✓ 11 feature gaps closed in IT402 Deliverable 2, a billing engine, and English/Shona/Arabic across every screen and document
- ✓ Runs fully offline — zero paid external accounts required to build or test

“A claim in this project should be a fact about the running system, not a target for it.”

Future Work

Beyond IT402:

1

Native-speaker review

Have Arabic and Shona speakers check the wording, and extend Shona beyond the main screens

2

Mobile-optimized tenant portal

Tenants are the role most likely to use the system from a phone

3

Live production use

render.yaml deploys to Render; real tenants still need S3 uploads and a long-term database plan

4

Live third-party verification

End-to-end tests against real Stripe, SendGrid, and Twilio sandbox accounts

5

Two-factor sign-in

A second factor for Owner, Manager and Admin accounts

Thank You

Questions?